

Hello everyone. This is Part 3 of BLA 2 in my AWS Machine Learning Engineer Associate MLA-C02 learning series. Part 1 covered collecting and storing data, and Part 2 explained transformation, feature engineering, embeddings, and RAG preparation. This final presentation focuses on whether the prepared data can actually be trusted.

I will begin with the main dimensions of data quality, including completeness, validity, consistency, and freshness. I will then explain how missing values, duplicate records, outliers, and invalid values require different cleaning decisions. The next topics are labeling and annotation, class imbalance, and the different points where bias can enter a dataset. I will finish by discussing integrity and safety checks for traditional machine learning and generative AI training data.

Data quality has several dimensions, and a dataset can succeed in one area while failing in another. Completeness asks whether required records and fields are present.

Validity asks whether values follow expected types, ranges, formats, and business rules. An age below zero, an impossible date, or a category outside the allowed list may indicate a quality problem.

Freshness asks whether the data arrives soon enough to represent current conditions. A model trained with old behavior may perform poorly after products, users, or policies change.

AWS Glue Data Quality can evaluate defined rules and report results, but a service cannot decide which rules matter to the business.

Data cleaning is not simply removing anything unusual. Missing values, duplicates, outliers, and invalid values can have different meanings. A missing value may occur because a field is optional, a sensor failed, a customer chose not to answer, or the information was unavailable at that moment. The team may remove the record, impute a value, add a missing indicator, or treat missing as a separate category. The choice depends on the reason and the amount of affected data.

A duplicate may come from a retry or repeated file delivery. It may also represent two valid events that happen to look similar.

An outlier can be a data-entry error, a broken sensor, a rare but valid example, or the exact event the model needs to detect.

That creates traceability and makes the resulting dataset reproducible.

Supervised learning requires labels that describe the desired outcome. Examples include fraud or legitimate, positive or negative sentiment, and defective or acceptable. Generative

AI preparation may also use prompt-response pairs, rankings, safety categories, and human preference data.

A reliable process begins with definitions. Labeling guidelines should explain each category, include positive and negative examples, and describe difficult edge cases. Labels may come from employees, external reviewers, subject-matter experts, automated methods, or a combination of these approaches.

Class imbalance occurs when some outcomes appear much more often than others. A fraud dataset may contain a very small number of fraudulent transactions. A manufacturing dataset may contain many normal products and only a few defects. In this situation, overall accuracy can be misleading. A model that predicts the majority class every time may report a high accuracy while completely failing on the rare class that matters.

Bias can enter a machine learning system at several points. Collection bias appears when the sample does not represent the intended population or environment. A dataset may exclude certain locations, languages, devices, age groups, or operating conditions. A model trained on that sample may perform well for represented groups and poorly for others.

Training data integrity means the dataset is trustworthy, traceable, permitted, and appropriate for the intended model. For traditional machine learning, checks include schema consistency, label quality, duplicate records, source permissions, and representative sampling.

To conclude Part 3 and BLA 2, trustworthy models begin with trustworthy data. Data quality should be measured through rules connected to the actual use case. Completeness, validity, consistency, freshness, uniqueness, and accuracy describe different risks, so one successful check does not guarantee overall quality.

Thank you for watching.